array

⇒ int age;

int a;
int b;
int c;
int d;
int e;

int a, b, c, d, e, f;

int a, b, c, d, e, f, --- z, aa, ab;

---

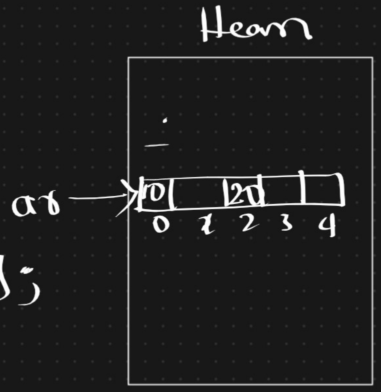
array :

[ ]

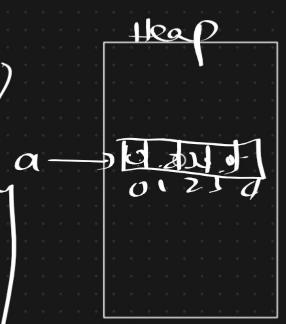
⇒ treated as an Object

⇒ Heap area

⇒ 5 students :-




```
int[] ar = new int[5];

int ar[] = new int[5];

ar[0] = 10;     ar[2] = 20;

S.o.p(ar[2]); 20;
```




```
int[] a = new int[5];

a[0] = 10;
a[1] = 20;
a[2] = 30;
a[3] = 40

int[] a = {10, 20, 30, 40};
```

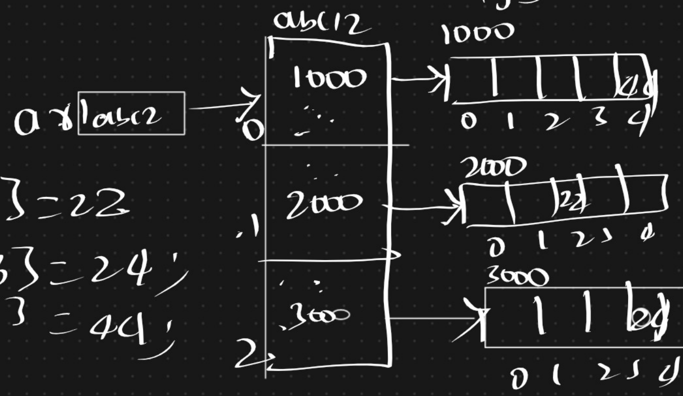
5 student in 3 d/f classrooms.

classes	students
0	⇒ 5
1	⇒ 5
2	⇒ 5

class student

```
int [][] ar = new int[3][5];
```

(i) (j)



```
ar[1][2] = 22
ar[2][3] = 24;
ar[0][4] = 44;
```

⇒ 2D Regular array

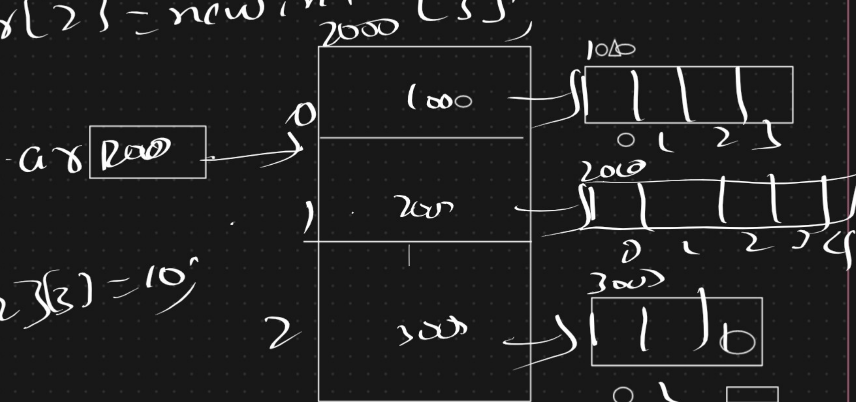
ar.length ⇒ 3

ar[i].length ⇒ 5

ar[1].length ⇒ 5

classes	student
0	4
1	5
2	3

```
int[][] ar = new int[3][];
ar[0] = new int[4];
ar[1] = new int[5];
ar[2] = new int [3];
```



```
ar[2][3] = 10;
```

2 Jagged array

ar[i].length

school	classes	student
0	0	4
	1	4
	2	4
1	0	
	1	4
	2	4

```
int[][][] ar = new int[2][3][4];
```

school	classes	students
0	0	4
	1	3
1	0	5
	1	3
	2	2

```
int[][][] ar = new int[2][][];
ar[0] = new int[2][];
ar[1] = new int[3][];

ar[0][0] = new int[4];
ar[0][1] = new int[3];

ar[1][0] = new int[5];
ar[1][1] = new int[3];
ar[1][2] = new int[2];
```

3D Jagged array: irregular

=> C

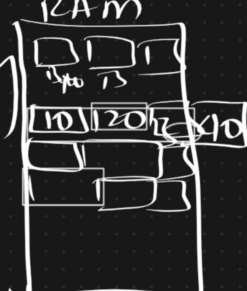
```
int[] ar;
ar[0]=10
ar[1]=20
ar[2]=30
ar[3]=40
```

{ BufferOverRun problem

4->4

4×4=16

RAM



int []ar = new int[3];

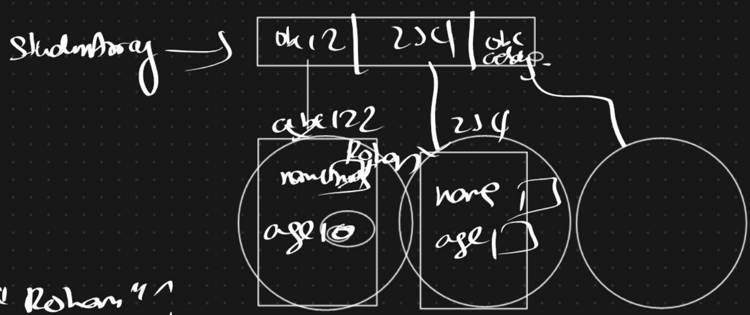
ar[0]=10;

ar[1]=20;

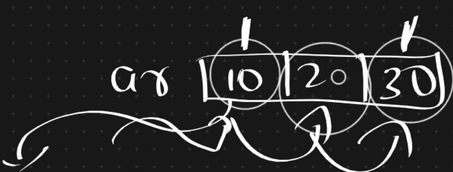
ar[2]=30;

ar[3]=40;

```java
class Student
{
    String name;
    int age;

}
public class LaunchAR4 {

    public static void main(String[] args) {
        Student []studentArray=new Student[3];

        studentArray[0]=new Student();
        studentArray[1]=new Student();
        studentArray[2]=new Student();
    }

}
```



studentArray[0].name = "Rohan";



for (int a : ar)

{

10

4 x 3 => 12 Bytes

}

RAM

int ar[] = new int[3];

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